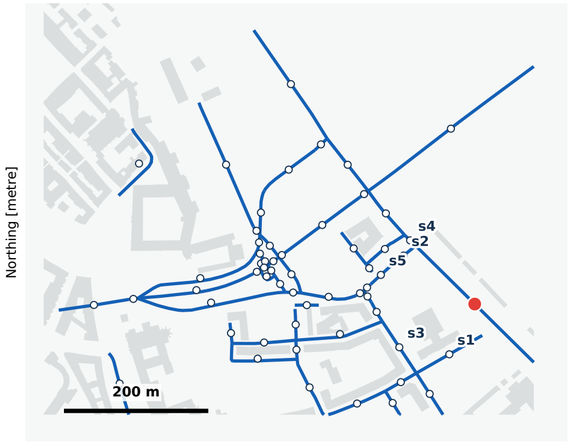
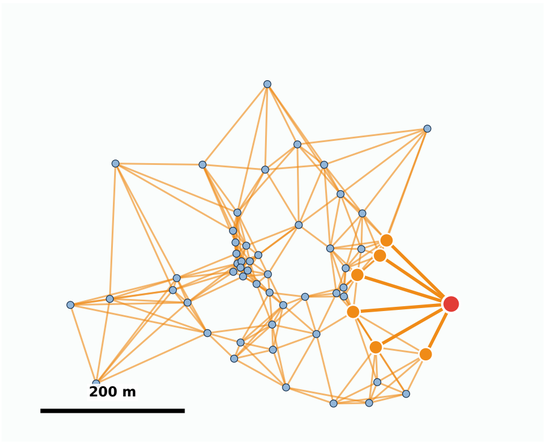
## (a) Input road network (source geometry)




## (b) Zoomed road segments and centroids




## (c) Graph construction (kNN ≤ 250 m)




## (d) Spatial neighborhood example




## (e) Edge construction procedure

**1. Compute segment centroids**

For each road segment $s_i$, compute $c_i$.

**2. Find spatial neighbors**

Select k=6 nearest centroids within 250 m.

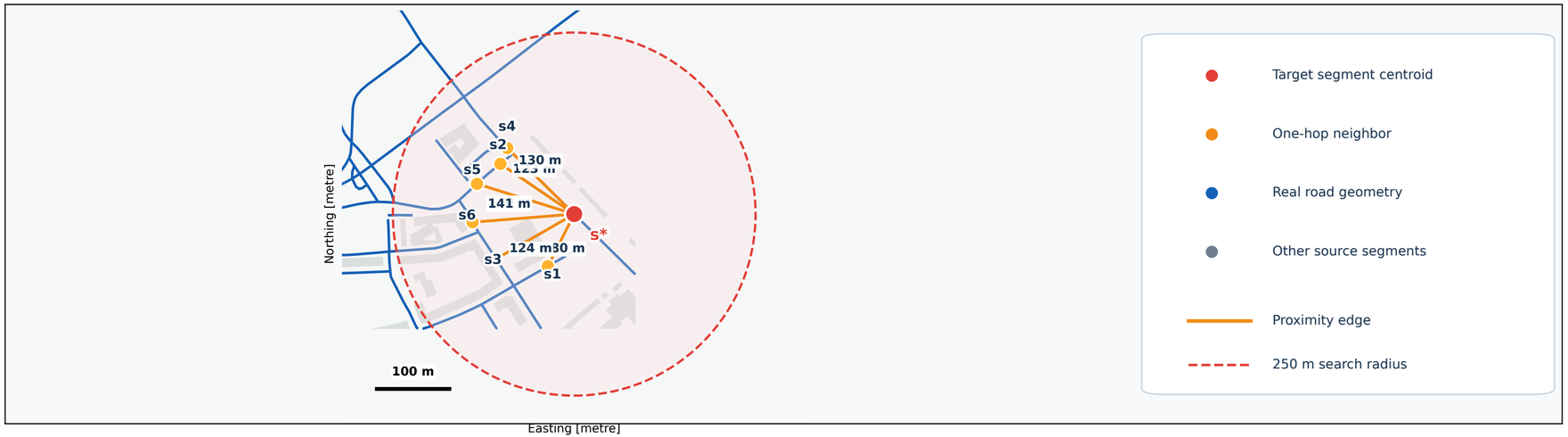
**3. Create undirected edges**

Add (i,j) if either centroid selects the other.

**4. Attach node attributes**

Combine road geometry and graph-context features.

## (f) Distance-based neighborhood illustration




All roads, buildings, centroid positions, and distances derive from the real GIS scene. Proximity edges do not imply physical road connectivity.